

with fuel to produce reformed fuel which flows over the anode side.

Next, the chemical reaction begins in the fuel cell. As the reformed fuel crosses the anode, it attracts oxygen ions from the cathode. The oxygen ions combine with the reformed fuel to produce electricity, water and small amounts of carbon dioxide.

The water gets recycled to produce the steam needed to reform the fuel. The process also generates the heat required by the fuel cell. As long as there is fuel, air and heat, the process continues producing clean, reliable and affordable energy.

One of the biggest problems with SOFCs is their temperature requirement—the ceramic squares become active only at extremely high temperatures (up to 1800°F). That means Bloom's cells will have to prove that they can remain durable under the stress. In general, Bloom expects that its fuel cell stacks will have to be switched out twice during the device's

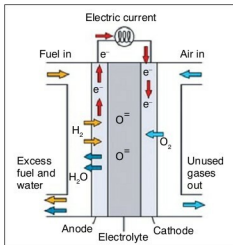


Fig. 4: Energy-generating process of Bloom's device

ten-year lifespan.

Bloom's device generates electricity at 50 per cent to 55 per cent conversion efficiency. In comparison, a solar system generally produces power at 10 to 15 per cent efficiency. But unlike solar panels, the Bloom Energy server produces CO<sub>2</sub> as a by-product.

According to the Energy Collective, "CO<sub>2</sub> emissions, when running on natu-

ral gas, would be just under 0.8 lbs/kWh, which compares favourably to electricity from central station coal-fired plants (2 lbs/kWh) or natural gas plants (roughly 1.3 lbs/kWh) and the national average for on-grid electricity (around 1.3-1.5 lbs/kWh)." If the box runs on landfill gas or biogas, it produces net zero carbon emissions.

Presently, the bloom energy is used in many parts of the world. Its disadvantages include large space occupancy and installation costs. But the technological developments are reducing these disadvantages.

### Inside the Bloom Box...

Inside the Bloom Box we have the fuel cell stack, fuel processor, an inverter and inlets for oxygen from air and natural gas and outlet for heat produced as a result of the chemical reaction. As the fuel and oxygen enter the Bloom Box, they convert into heat and electricity.

If bio-gas is used instead of natural gas as a fuel, the carbon dioxide pro-

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